

EE Simulation Scenarios Results

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Date: 2026-06-03

Scope and data used

This note summarizes simulated outcomes from CSV outputs in ExcelFiles/Output/ for:

- EE_PDP8
- EE_Directive10
- EE_Directive10_NoBESS
- EE_PDP8_PV_BESS_NoBESS

Requested-but-missing output files at the time of reporting:

- EE_Directive10_PV_BESS.csv (not found)
- EE_PDP8_PV_BESS.csv (not found)

Metric definitions

- GDP growth: annual growth of Y_1 in percent.
- GDP level: model level Y_1 (reported relatively in comparison tables).
- Investment share: $I_1 / Y_1 * 100$.
- Consumption share: $C_1 / Y_1 * 100$.
- Energy intensity index: $((Q_{A_2_1} + Q_{PV_1}) / (Q_{A_2_1}(2026) + Q_{PV_1}(2026))) / (Y_1 / Y_1(2026)) * 100$.
- Final energy demand index: $((Q_{A_F_2_1} + Q_{PV_1}) / (Q_{A_F_2_1}(2026) + Q_{PV_1}(2026))) * 100$.
- Emissions index: $E_1 / E_1(2026) * 100$.

Headline findings

1. EE_Directive10 and EE_Directive10_NoBESS deliver slightly higher GDP levels than EE_PDP8 throughout the horizon (about +0.44% to +0.58% by 2030-2050).
2. The two Directive10 variants are very close to each other in macro aggregates, indicating limited additional macro divergence from BESS in the available run set.
3. Energy intensity improves (lower index) in Directive10 variants relative to EE_PDP8.
4. Emissions are identical across the four available EE outputs at the key years and over the 2026-2050 cumulative index sum.

Key-year levels

2030

Scenario	GDP growth (%)	Investment share (% GDP)	Consumption share (% GDP)	Energy intensity (index)	Final energy demand (index)	Emissions (index)
EE_PDP8	9.972	36.290	52.233	80.135	222.938	111.416
EE_Directive10	9.963	36.494	52.146	78.256	227.025	111.416
EE_Directive10_NoBESS	9.963	36.492	52.147	78.255	227.007	111.416
EE_PDP8_PV_BESS_NoBESS	9.971	36.388	52.187	79.200	224.930	111.416

2040

Scenario	GDP growth (%)	Investment share (% GDP)	Consumption share (% GDP)	Energy intensity (index)	Final energy demand (index)	Emissions (index)
EE_PDP8	7.503	35.597	45.843	53.952	256.037	62.988
EE_Directive10	7.505	35.689	45.802	52.316	261.033	62.988
EE_Directive10_NoBESS	7.504	35.684	45.805	52.317	260.907	62.988
EE_PDP8_PV_BESS_NoBESS	7.501	35.636	45.827	53.204	258.204	62.988

2050

Scenario	GDP growth (%)	Investment share (% GDP)	Consumption share (% GDP)	Energy intensity (index)	Final energy demand (index)	Emissions (index)
EE_PDP8	7.553	44.870	39.579	21.915	203.112	25.000
EE_Directive10	7.433	44.852	39.641	21.065	204.412	25.000
EE_Directive10_NoBESS	7.455	44.860	39.632	21.083	204.313	25.000
EE_PDP8_PV_BESS_NoBESS	7.508	44.865	39.604	21.540	203.639	25.000

Differences vs EE_PDP8

2030 deltas (scenario minus EE_PDP8)

Scenario	GDP growth (pp)	GDP level (%)	Investment share (pp GDP)	Consumption share (pp GDP)	Energy intensity (percentage points)	Final energy demand (percentage points)
EE_Directive10	-0.009	+0.442	+0.205	-0.087	-1.880	+4.087
EE_Directive10_NoBESS	-0.009	+0.440	+0.203	-0.086	-1.880	+4.069
EE_PDP8_PV_BESS_NoBESS	-0.001	+0.219	+0.098	-0.046	-0.936	+1.992

2040 deltas (scenario minus EE_PDP8)

Scenario	GDP growth (pp)	GDP level (%)	Investment share (pp GDP)	Consumption share (pp GDP)	Energy intensity (percentage points)	Final energy demand (percentage points)
EE_Directive10	+0.002	+0.579	+0.092	-0.041	-1.636	+4.995
EE_Directive10_NoBESS	+0.001	+0.571	+0.086	-0.038	-1.635	+4.869
EE_PDP8_PV_BESS_NoBESS	-0.001	+0.262	+0.038	-0.016	-0.748	+2.166

2050 deltas (scenario minus EE_PDP8)

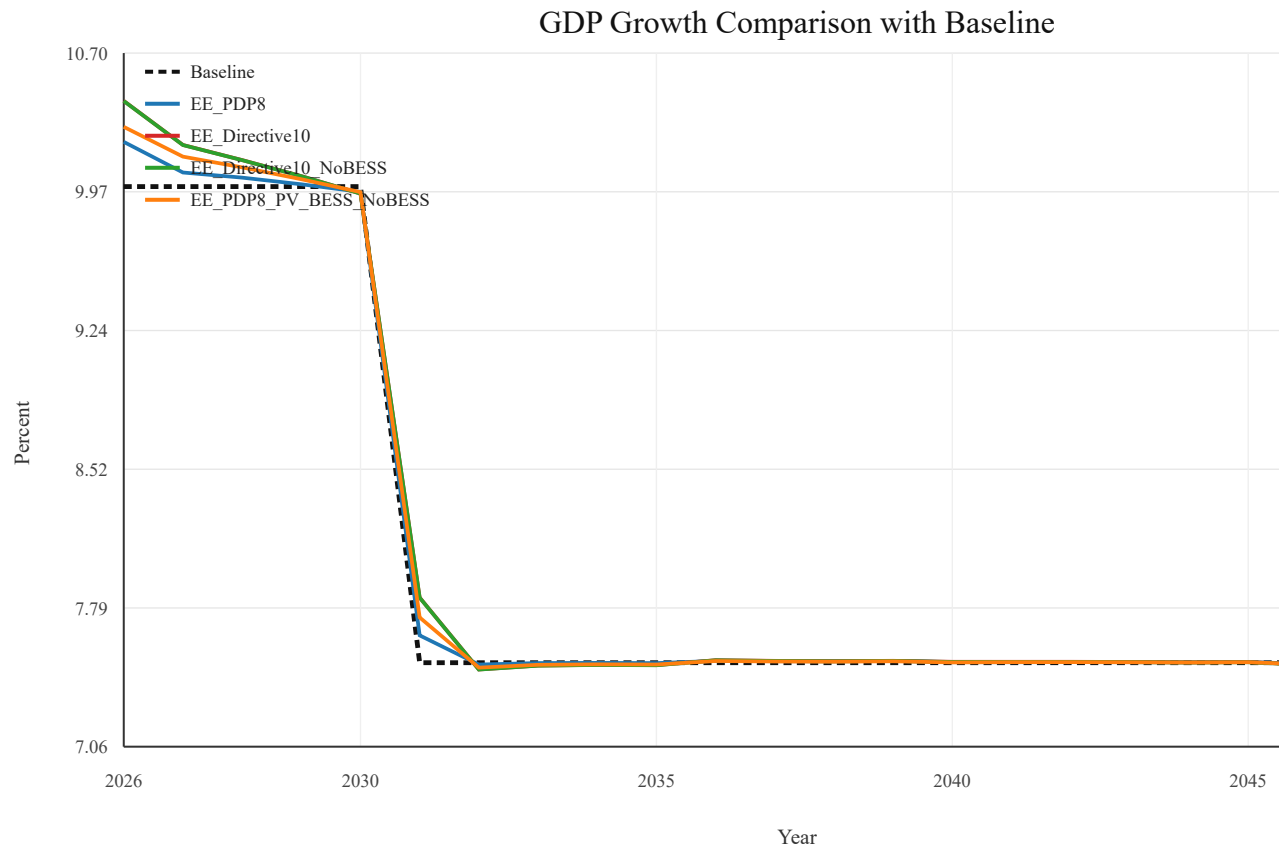
Scenario	GDP growth (pp)	GDP level (%)	Investment share (pp GDP)	Consumption share (pp GDP)	Energy intensity (percentage points)	Final energy demand (percentage points)
EE_Directive10	-0.120	+0.446	-0.018	+0.061	-0.850	+1.300
EE_Directive10_NoBESS	-0.098	+0.459	-0.011	+0.053	-0.833	+1.201
EE_PDP8_PV_BESS_NoBESS	-0.044	+0.207	-0.005	+0.025	-0.376	+0.527

Aggregate indicators (2026-2050)

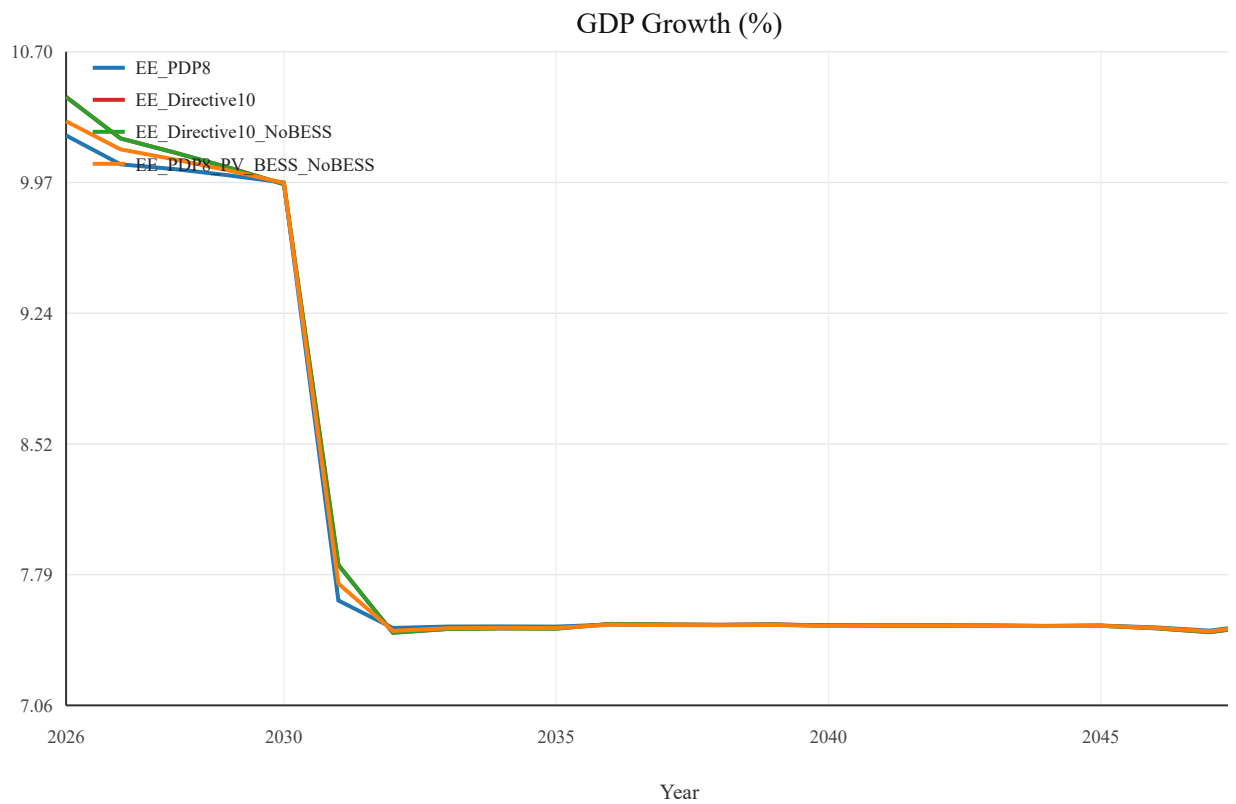
Scenario	Average GDP growth (%)	Sum of emissions index
EE_PDP8	8.015	1792.148
EE_Directive10	8.035	1792.148
EE_Directive10_NoBESS	8.035	1792.148
EE_PDP8_PV_BESS_NoBESS	8.024	1792.148

Plots

GDP Growth Comparison with Baseline

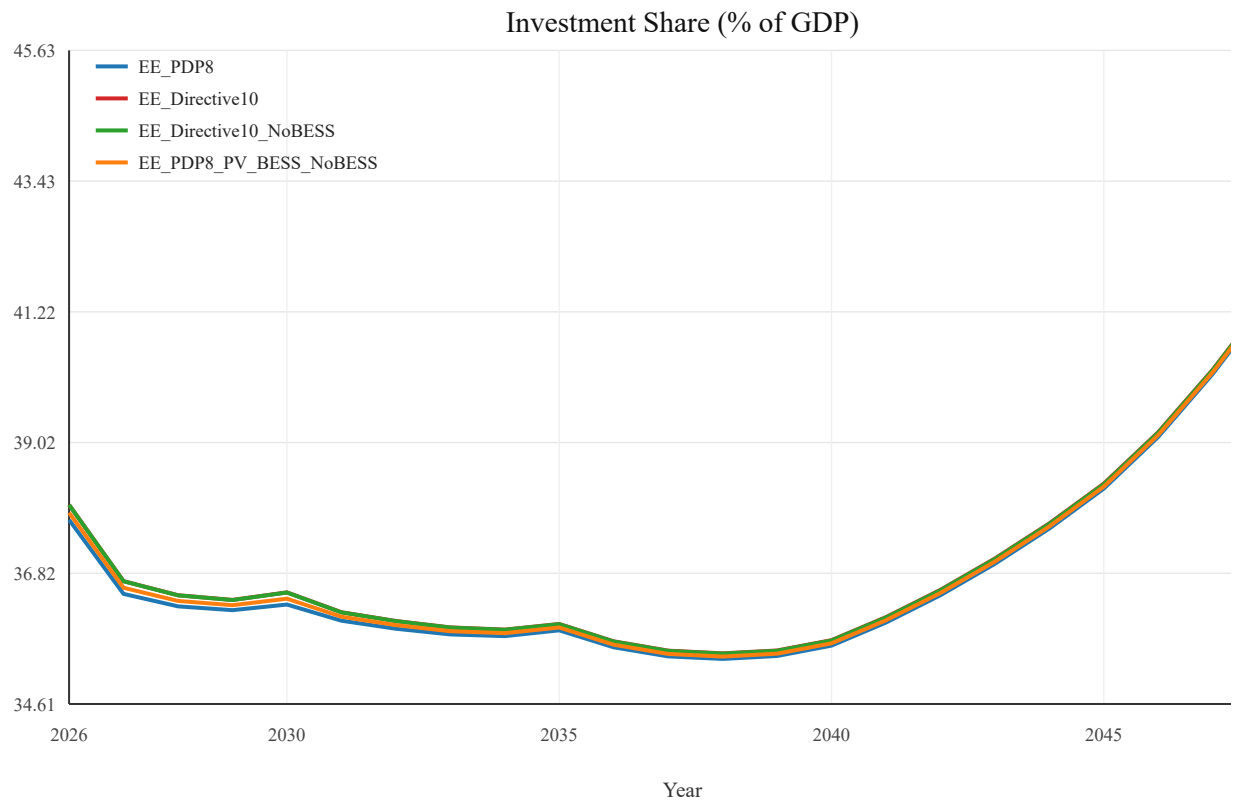


GDP Growth Comparison with Baseline

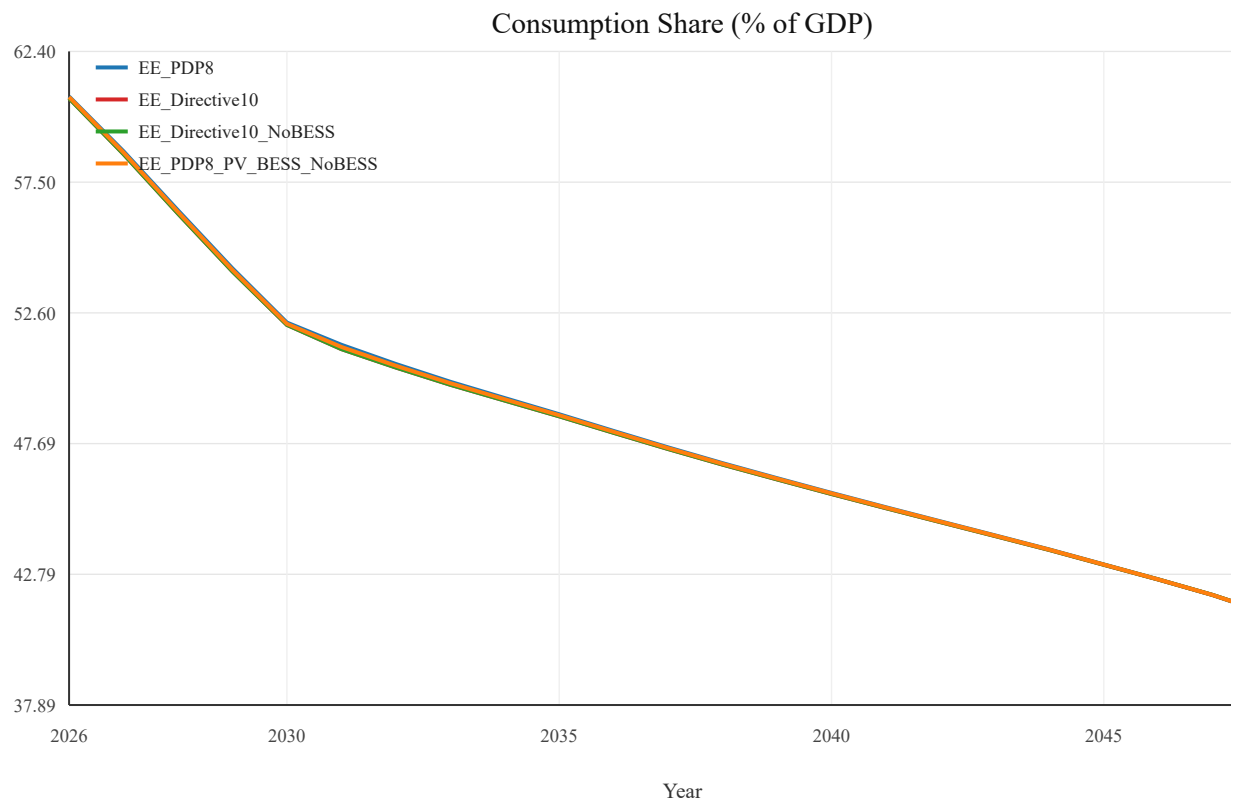
GDP Growth

GDP Growth

Investment Share of GDP

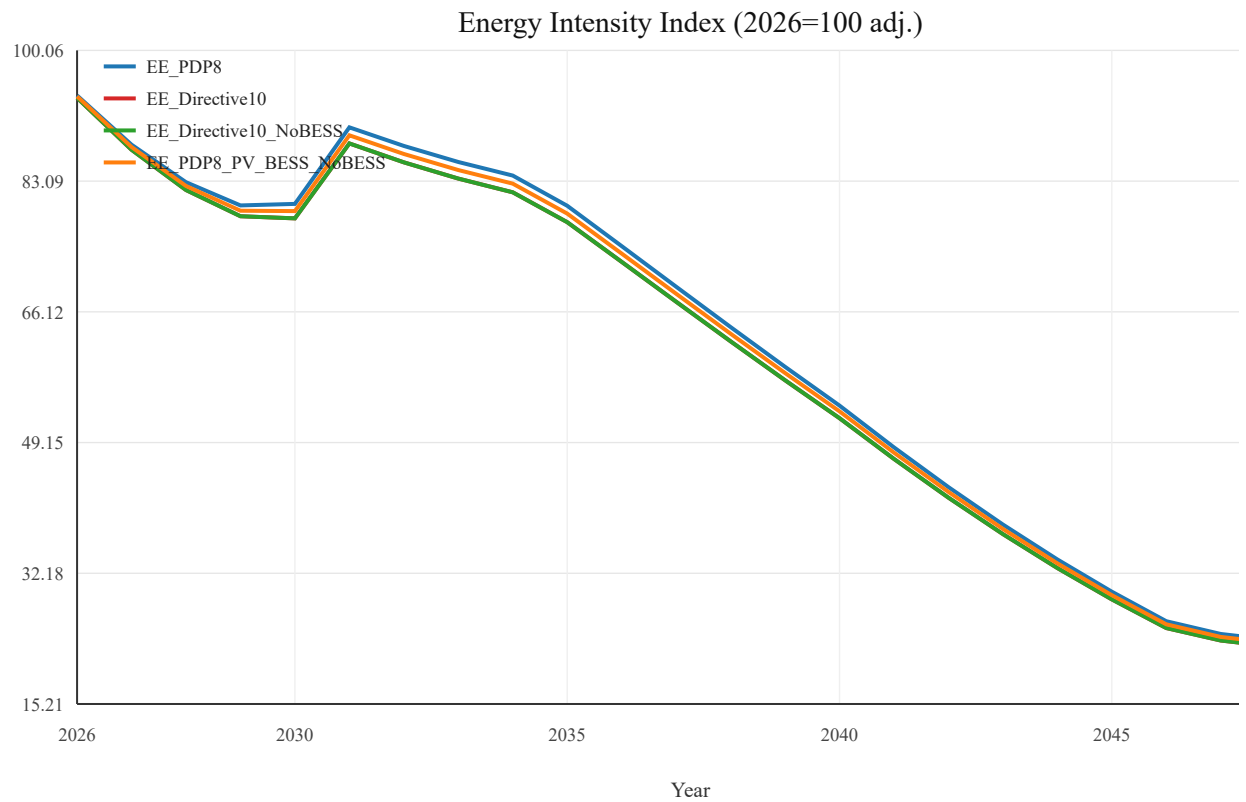


Investment Share of GDP

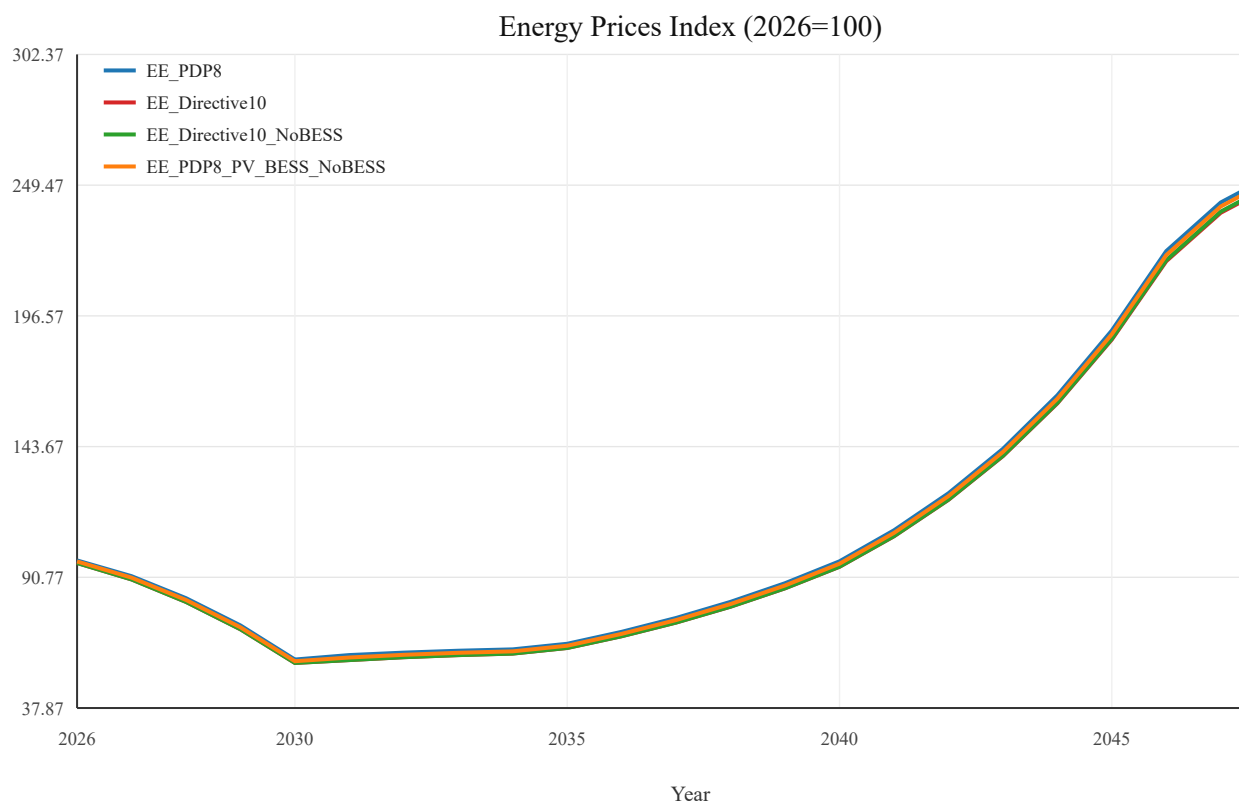
Consumption Share of GDP

Consumption Share of GDP

Energy Intensity Index

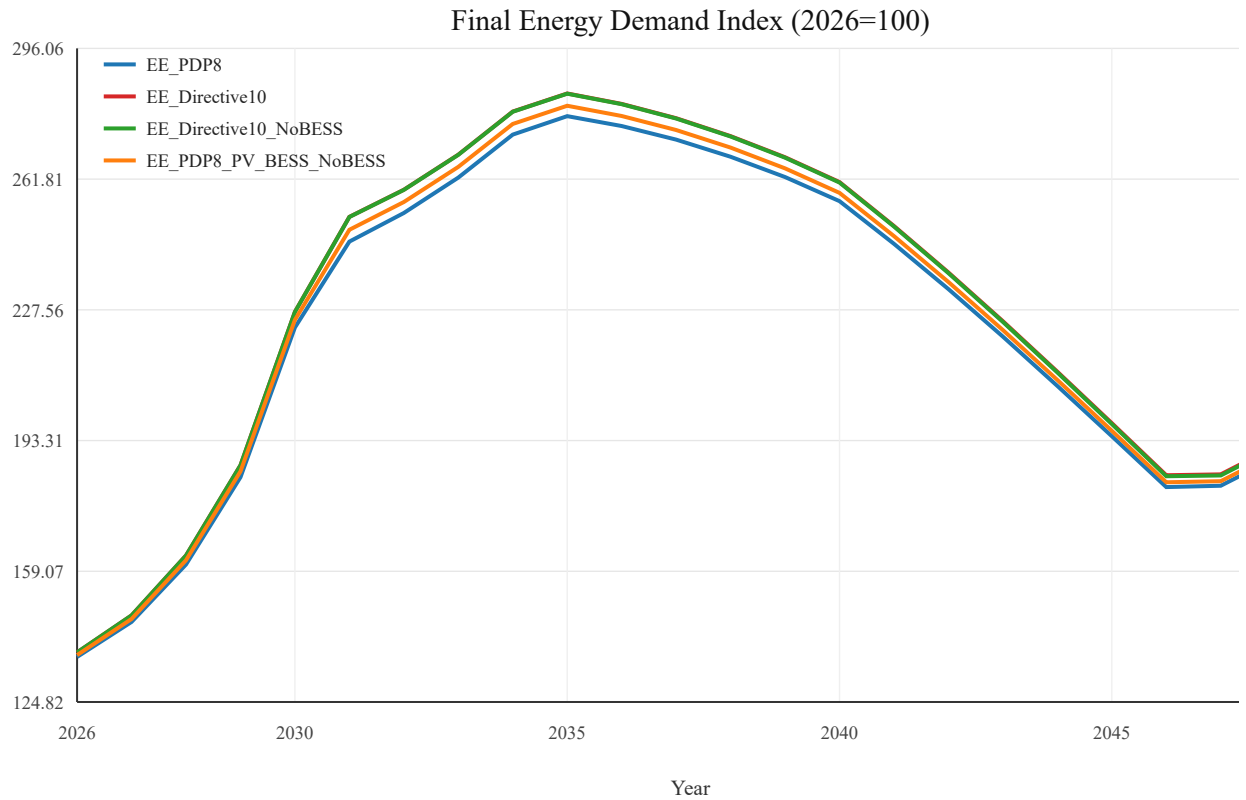


Energy Intensity Index

Energy Prices Index

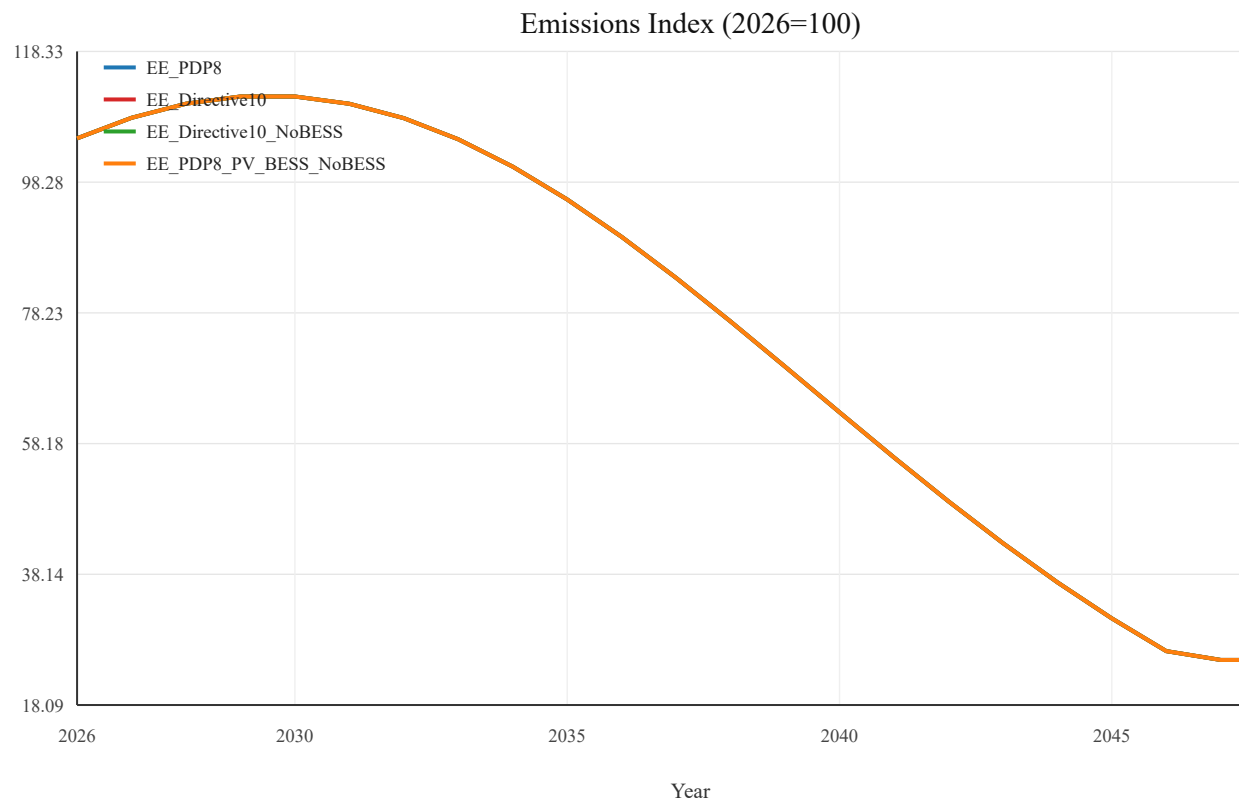
Energy Prices Index

Final Energy Demand Index



Final Energy Demand Index

Emissions Index

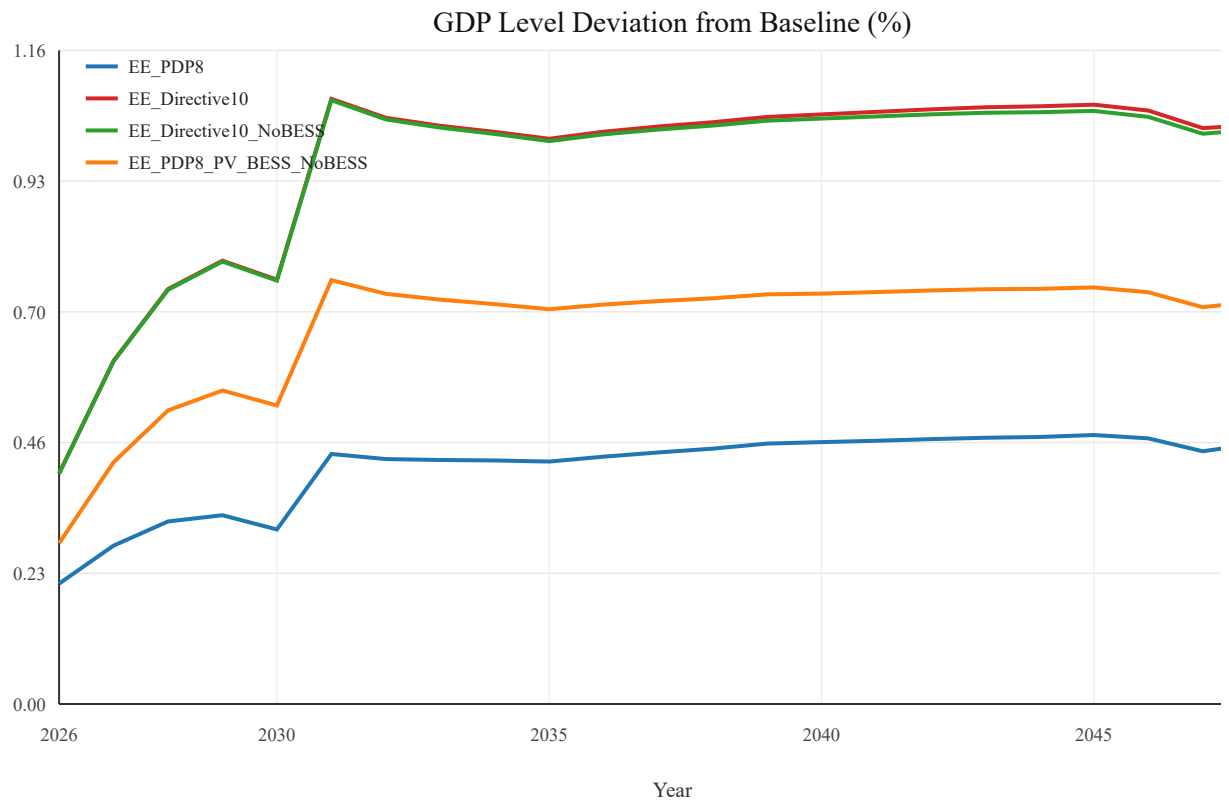


Emissions Index

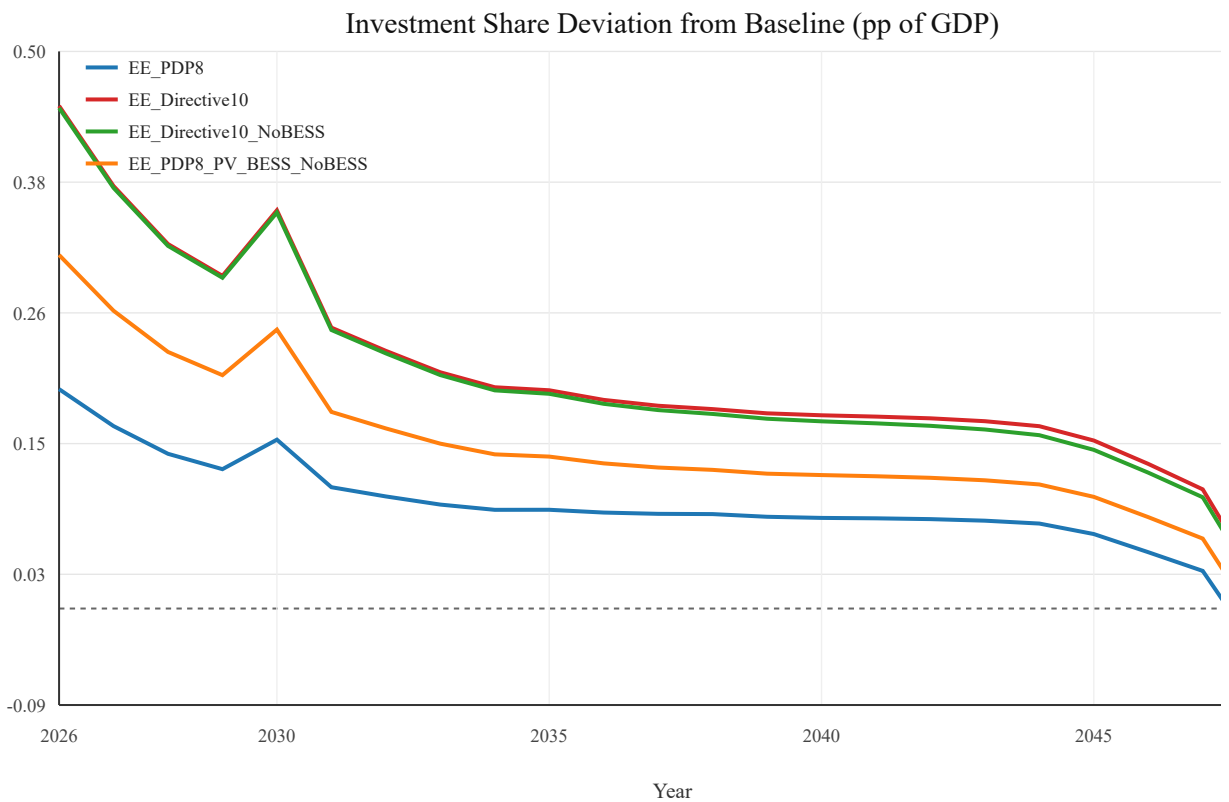
Deviations from Baseline Plots

These charts show scenario differences relative to Baseline directly (zero line = same as Baseline).

GDP Level Deviation from Baseline

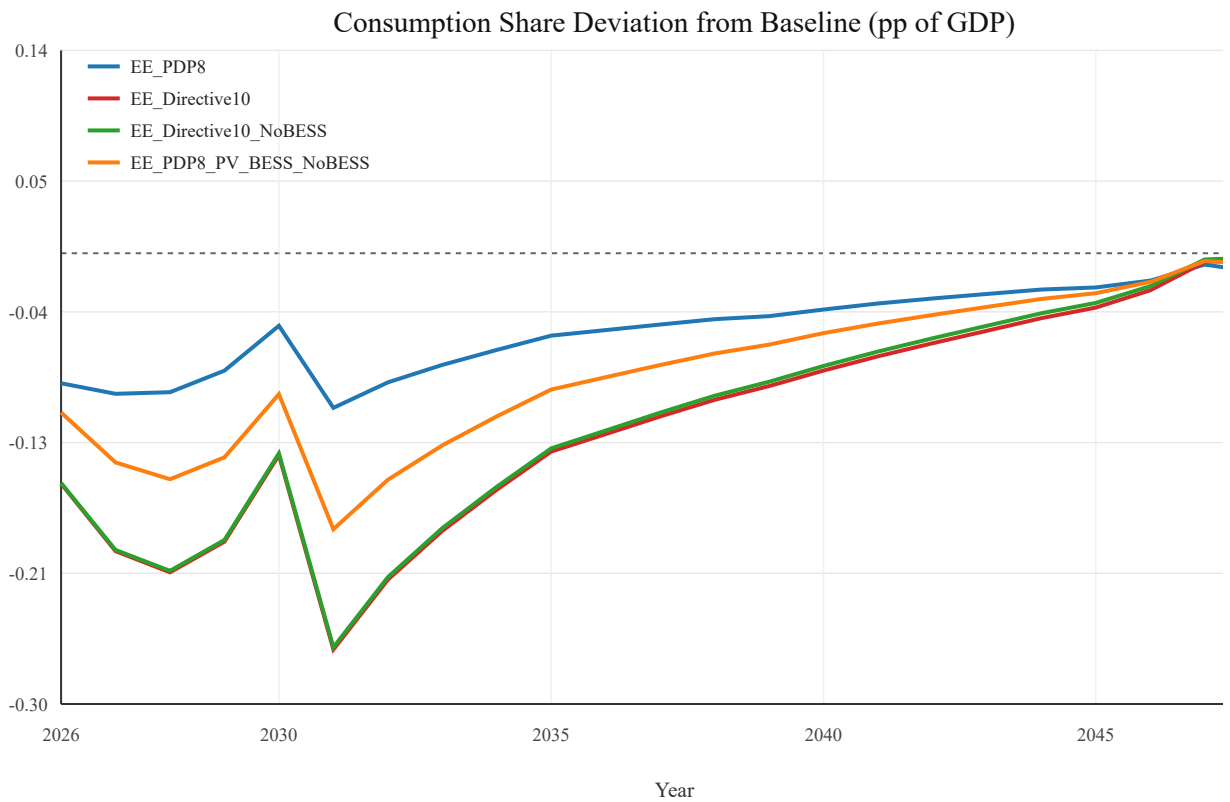


GDP Level Deviation from Baseline

Investment Share Deviation from Baseline

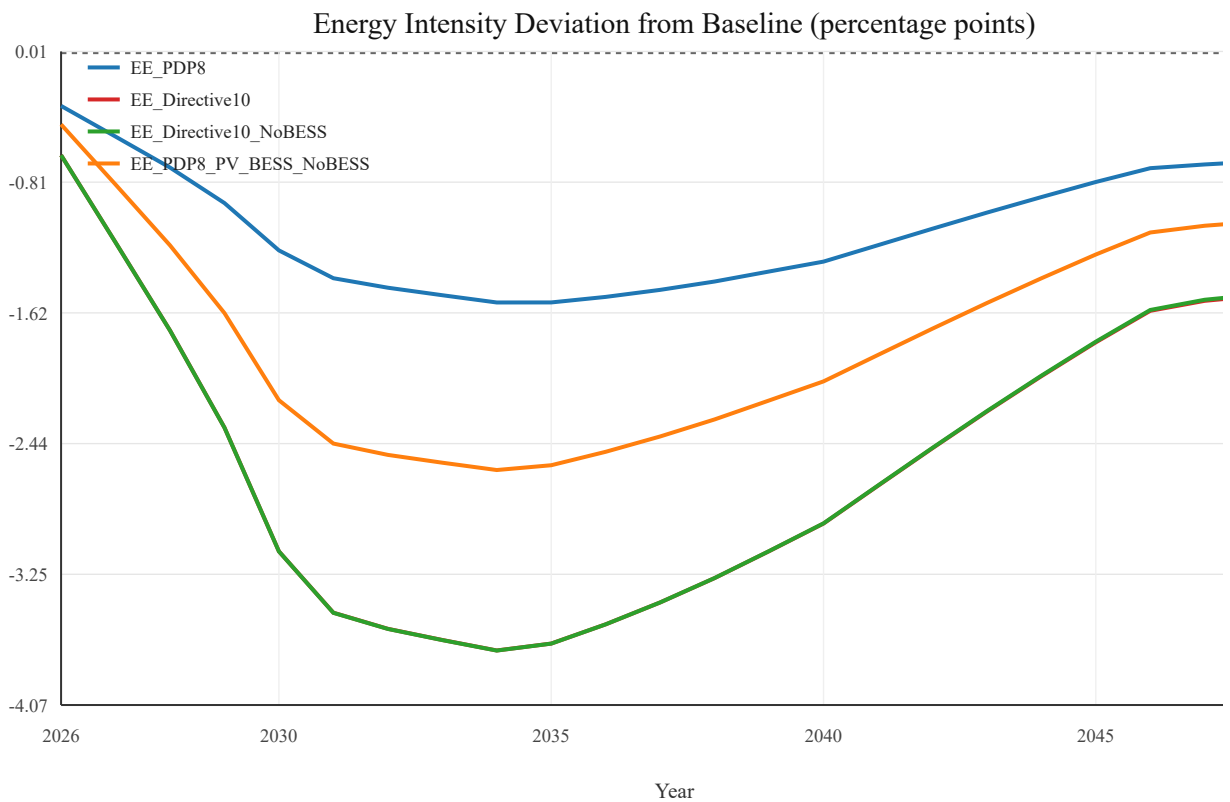
Investment Share Deviation from Baseline

Consumption Share Deviation from Baseline



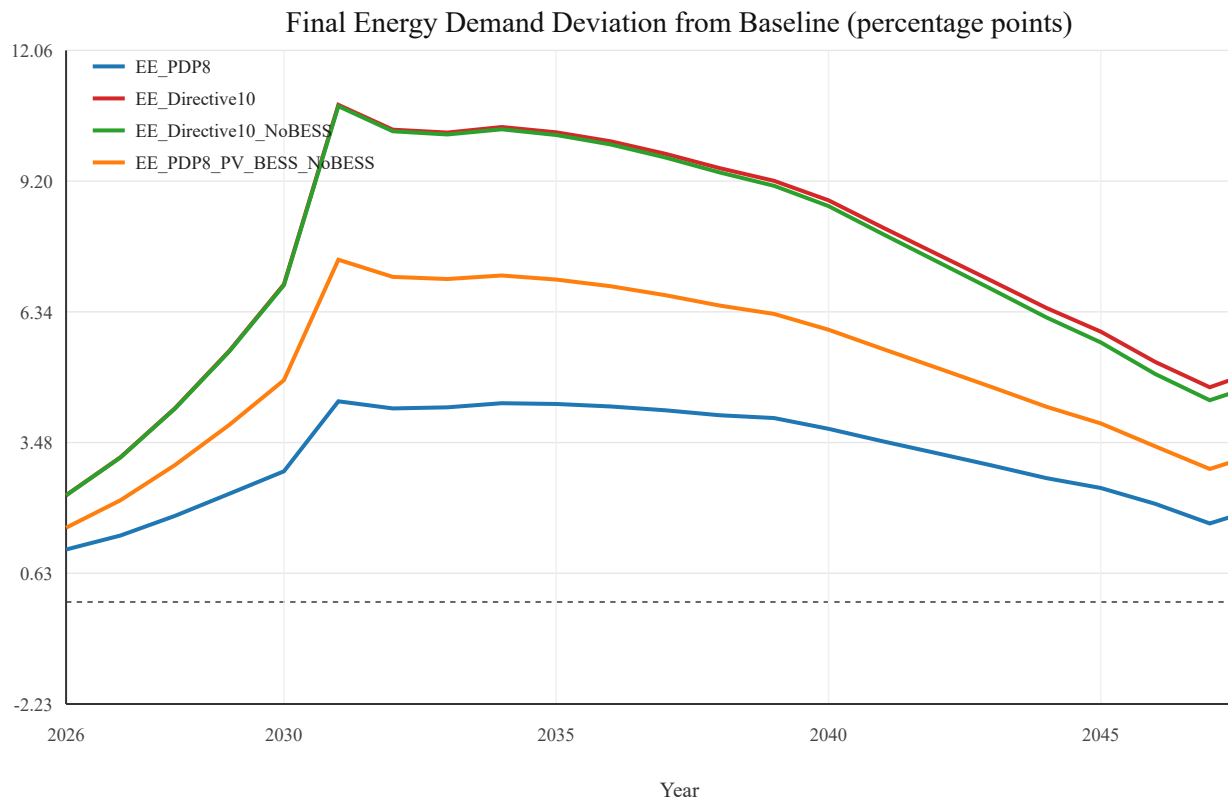
Consumption Share Deviation from Baseline

Energy Intensity Deviation from Baseline



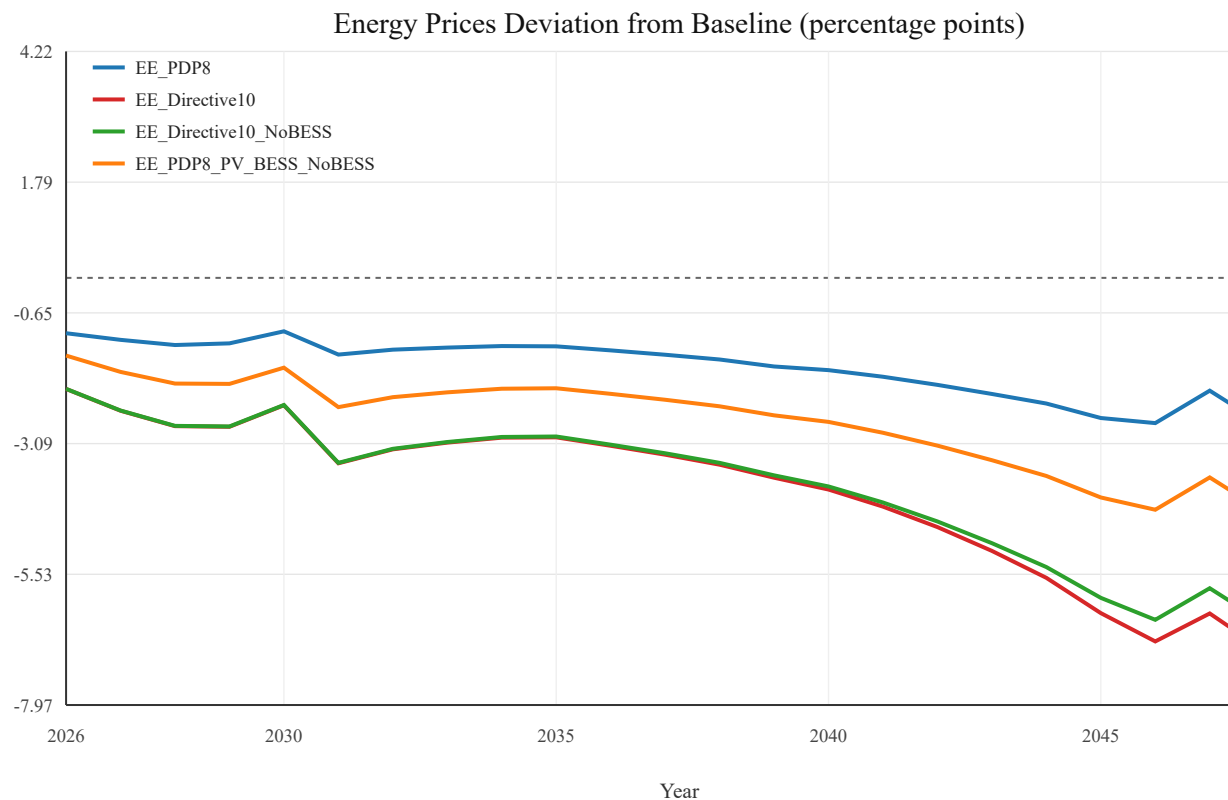
Energy Intensity Deviation from Baseline

Final Energy Demand Deviation from Baseline



Final Energy Demand Deviation from Baseline

Energy Prices Deviation from Baseline



Energy Prices Deviation from Baseline

Interpretation notes

- The near-zero gap between EE_Directive10 and EE_Directive10_NoBESS in these outputs suggests BESS effects are modest for the selected macro aggregates under the current calibration/run settings.
- Identical emissions paths indicate either a shared binding constraint/policy path in these runs or scenario differences that primarily affect non-emissions channels.

- To complete the originally requested five-scenario comparison, generate and export the missing PV+BESS scenario outputs (EE_Directive10_PV_BESS.csv and/or EE_PDP8_PV_BESS.csv) and append them to this table set.